

8.7 Classwork: Function Composition — Markscheme

1. Functions f and g given by the table.

(a) $f(1) = 5.$ A1

(b) $(g \circ f)(1) = g(f(1)) = g(5)$ M1

$= 3.$ A1

(c) $(f \circ g)(-2) = f(g(-2)) = f(1)$ M1

$= 5.$ A1

(d) $f^{-1}(-2) = 3$ (since $f(3) = -2$). A1

2. $f(x) = 2x - 1$, $g(x) = x^2 + 3$.

(a) $(f \circ g)(x) = 2(x^2 + 3) - 1$ M1

$= 2x^2 + 5.$ A1

(b) $(g \circ f)(x) = (2x - 1)^2 + 3 = 4x^2 - 4x + 1 + 3$ M1

$= 4x^2 - 4x + 4.$ A1

(c) $(g \circ f)(-1) = g(-3) = 9 + 3 = 12.$ A1

Or $4(-1)^2 - 4(-1) + 4 = 12.$

(d) $2x^2 + 5 = 13$, $2x^2 = 8$, $x^2 = 4$ M1

$x = \pm 2.$ A1

Award A0 if only $x = 2$ is given.

3. $f(x) = \frac{1}{x-2}$, $g(x) = x + 5$.

(a) $(f \circ g)(x) = \frac{1}{(x+5)-2}$ M1

$= \frac{1}{x+3}.$ A1

(b) $x = -3.$ A1

(c) $(g \circ f)(x) = \frac{1}{x-2} + 5 = \frac{1+5(x-2)}{x-2}$ M1

$$= \frac{5x - 9}{x - 2}.$$

A1

$$\text{(d)} \quad (f \circ f)(x) = \frac{1}{\frac{1}{x-2} - 2} = \frac{x-2}{1-2(x-2)}$$

M1

$$= \frac{x-2}{5-2x}.$$

A1

4. $f(x) = 3x - 4, \quad (f \circ g)(x) = 6x + 2.$

$$\text{(a)} \quad f(g(x)) = 3g(x) - 4 = 6x + 2$$

M1

$$3g(x) = 6x + 6$$

A1

$$g(x) = 2x + 2.$$

A1

(b) Let $y = 2x + 2$ and swap x and y : $x = 2y + 2$

M1

$$y = \frac{x-2}{2}$$

A1

$$g^{-1}(x) = \frac{x-2}{2}.$$

A1